
Project Recommendation 1 -

LogiSecure

On-Prem Logistics AI-Copilot

Problem Statement

Global companies rely on real-time information regarding their land, sea, and air assets to maintain their competitive edge. However, current solutions force a choice between two critical risks: relying on public clouds that expose highly confidential operational data to industrial espionage, or isolating themselves within local systems disconnected from the dynamic global context (weather, traffic, and geopolitics).

This operational friction leads to costly delays, logistical inefficiencies, and a significant loss of the ability to make proactive, automated supply chain decisions.

Proposed Solution

LogiSecure AI is an enterprise-grade platform powered by autonomous agents designed to automate real-time domestic and international logistics, featuring 100% on-premise execution via AMD ROCm for maximum data confidentiality. It functions as an agentic control hub that integrates global traffic data to autonomously optimize routes, mitigate supply chain disruptions, and displays global supply chain information for suppliers & customers.

Workflow

Step 1 – Global Supply Chain Monitoring

The system continuously listens to open-source and global APIs (GNSS/GPS, Baidu Maps, aviation, and maritime transponders) to map live transit routes.

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Step 2 – Incident Detection

A global anomaly is registered externally (e.g., a port strike, extreme weather typhoon, or major highway bottleneck).

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Step 3 – Local Asset Correlation

The **LogiSecure AI** agent queries the secure, on-premise local database to identify exactly which confidential shipments, cargo manifests, and corporate assets are affected by the incident.

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Step 4 – Local AMD-Powered Inference

Using the **AMD ROCm** stack and local GPU/NPU hardware, an internal LLM analyzes the disruption data locally to simulate impact, calculate alternative multi-modal routes, and measure financial metrics.

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Step 5 – Agentic Autonomous Execution

The system generates a proactive response plan, automatically pushing the updated GPS coordinates to external transit operators via API and leaving a pre-drafted client communication alert ready for final approval.

Expected Output

The system generates:

- **Real-Time Global Risk Map Visuals:** A dynamic 3D dashboard displaying live vehicle positions overlaid with proactive risk alerts triggered by weather, traffic, or geopolitical events.
- **On-Premise Privacy Assurance Logs:** System telemetry reports confirming that 100% of confidential corporate asset data, manifests, and pricing structures remain securely processed inside the local AMD hardware without external cloud leakage.

- **Autonomous Multi-Modal Rerouting Suggestions:** Instant alternative transport route matrices (air, sea, or land) calculated by local models to minimize delays and financial losses during an active supply chain disruption.
 - **Automated API Fleet Adjustments:** Immediate data syncing payloads sent to external driver GPS apps and maritime/aviation transport systems via secure APIs to steer active shipments away from bottlenecks.
 - **Pre-Drafted B2B Client Notifications:** Tailored customer-facing status updates, custom delays summaries, and new Estimated Time of Arrival (ETA) email drafts ready for single-click executive approval.
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Advantages

- **Solves a Critical Global Logistics Pain Point:** Resolves the trade-off between real-time data visibility and corporate data leaks for multi-modal international freight.
 - **Demonstrates High-Impact Use of AMD Hardware:** Seamlessly integrates with the **AMD ROCm™** ecosystem, proving that heavy agentic workloads can run locally on **AMD Radeon™** or **Instinct™** GPUs.
 - **Drastically Reduces Recurrent Operational Costs:** Eliminates the ongoing token fees of third-party cloud AI vendors by centralizing inference on-premise, delivering immediate infrastructure ROI.
 - **Massive Enterprise B2B Market Potential:** Leverages an urgent corporate demand for data sovereignty in supply chains, opening a highly profitable licensing and support software model.
 - **Continuous Optimization via Multi-API Feedback:** The internal agent constantly refines its routing predictions and situational logic by learning from real-time global traffic, weather, and telemetry data streams.
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Challenges

- **API Data Heterogeneity & Real-Time Ingestion:** Consolidating massive, unstructured data streams from different international APIs (such as Western GPS, Eastern Baidu Maps, and maritime AIS) into a unified format in real-time requires a highly resilient data engineering pipeline.
- **On-Premise Hardware Dependencies:** Deploying the software *on-premise* means clients must have specific AMD hardware infrastructure (such as Radeon

or Instinct GPUs) or dedicated edge nodes already installed, which can slow down initial enterprise sales cycles.

- **Local Hardware Compute Constraints:** Running multi-modal AI agents locally forces the system to operate under strict hardware boundaries.
 - **Hybrid Network Security Execution:** Merging a 100% secure, offline internal corporate database with real-time outbound requests to public transit APIs demands a rigid network architecture to guarantee zero data leakage during external calls.
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Suggested Technology Stack

- **Frontend**
 - React.js
 - Tailwind CSS
 - Lucide React (for logistics/map iconography)
 - **Backend & Data Pipeline**
 - FastAPI
 - Python
 - Apache Kafka (for streaming real-time API telemetry)
 - PostgreSQL + PostGIS (for spatial asset tracking)
 - **AI & Agentic Framework**
 - LangGraph / AutoGen (for complex multi-agent routing workflows)
 - Llama-Index (for secure, local RAG on company data)
 - Llama 3.1 8B / 70B (quantized for edge inference)
 - Llama 3.2 Vision (for potential yard camera or video log analysis)
 - **Infrastructure & Hardware**
 - AMD ROCm™ Ecosystem
 - Docker containers (optimized for ROCm deployment)
 - AMD Radeon™ RX / PRO GPUs (for local client deployment)
 - AMD Instinct™ Accelerators / AMD Developer Cloud (for heavy fine-tuning and testing)
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Team Responsibilities

Team Member	Responsibility
Technical Lead	Overall architecture, database, task management
Frontend Developer	Interactive 3D tracking dashboard, maps integration, and alert interfaces
Backend Developer	Global API integrations (GNSS, Open-SkyNetwork, OpenStreetMap), Kafka pipelines, and database
Cloud AI Developer	Multi-agent orchestration, function calling tools, and local RAG architecture
Local AI Developer	Model optimization, quantization, and deployment using the AMD ROCm stack
DevOps & Presenter	On-premise Docker containerization, network security, and infrastructure setup. Business validation, pitch creation & demo presentation
